

Yusuke Ozaki

Kwansei Gakuin University, School of Engineering, Program of Computer Science
Exchange Student, Department of Computer Science, University at Albany, SUNY

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OBJECTIVE

- Seeking a Summer 2026 internship in machine learning and AI for science.
- Interested in problems at the intersection of bioengineering, active learning, and laboratory automation.
- I aim to contribute by implementing laboratory automation and developing active learning methods for the target system on top of that automation infrastructure.

EDUCATION

- **Kwansei Gakuin University** Apr. 2022 – Mar. 2027 (expected)
Bachelor of Engineering in Computer Science, School of Engineering Sanda, Hyogo, Japan
 - GPA: 3.62/4.00
- **University at Albany, SUNY** Aug. 2025 – May 2026 (expected)
Exchange Student, Department of Computer Science Albany, NY, USA
 - GPA: 4.00/4.00

RESEARCH INTERESTS

- AI for bioengineering and biology, including graph neural networks (GNNs) for molecular and cellular systems and gene regulatory network (GRN) inference from single-cell omics data.
- Design of active-learning (Bayesian optimization) frameworks that autonomously drive scientific experiments, with a focus on protocol-aware lab automation and data-efficient exploration of experimental conditions.

PRESENTATIONS / MANUSCRIPTS

- [1] Yusuke Ozaki, Chong Liu (Jan. 2026, *Under review*). **Cumulative Regret Analysis of Lipschitz Bandits with Lipschitz Propagation**. Short manuscript under review at the *UAI 2026*. OpenReview: [vC6ZQk2SiU](#).
- [2] Yusuke Ozaki, Dhaval C Patel (Jan. 2026, *Under review*). **SPIN: Structural LLM Planning via Iterative Navigation**. Short manuscript under review at the *ACL ARR 2026 January Submission*. OpenReview: [QmXiMRpBR0](#).
<https://openreview.net/pdf?id=7k2vOhYPnd>
- [3] Yusuke Ozaki (Sep. 2025). **Object-Flow Machine Learning: Active learning framework utilizing protocols information**. Poster presented in the Non-Archival Content Track at the *AutoML Conference 2025 (AutoML 2025)*, New York City, USA, Sep. 2025. OpenReview: [7k2vOhYPnd](#)
- [4] Yusuke Ozaki (May 2025). **Development of a machine learning method utilizing experimental protocols to predict outcomes and active learning framework employing the method**. Oral presentation at the *Student Symposium 2025, RIKEN Center for Biosystems Dynamics Research*, Kobe, Japan, May 2025.
- [5] Yusuke Ozaki (Feb. 2025). **Development of a machine learning method utilizing protocols to predict experimental outcomes**. Poster presentation at the *Spring School for Theoretical Biology 2025 (2025)*, Hiroshima University, Hiroshima, Japan, Feb. 2025.

EXPERIENCE

- **RIKEN Center for Biosystems Dynamics Research, Laboratory for Biologically Inspired Computing**  Jun 2024 – Aug 2025
Research Part-timer II Kobe, Japan
 - Worked in Koichi Takahashi's Laboratory for Biologically Inspired Computing on AI- and robotics-driven automation of wet-lab experiments.
 - Developed machine-learning based experimental simulators and protocol-aware active learning workflows for optimizing high-throughput experiments using automated liquid-handling systems.
 - Contributed to Python codebases and data-processing pipelines for integrating experiment logs, protocol representations, and optimization results.
- **Yachie Laboratory**  Feb 2025 – Aug 2025
Research Assistant Osaka Japan
 - Assisted in computational projects on gene regulatory network inference from single-cell omics data, combining transcriptomic and chromatin-accessibility measurements.

- Built and maintained analysis pipelines for single-cell RNA-seq and ATAC-seq using Python-based tools and graph/network modeling methods.
- Collaborated with experimental biologists to interpret network models and compare alternative GRN-inference workflows.
- **Metadata Incorporated**  Apr 2024 – Jun 2024
Software Engineer (Part-time), Web Application Development Remote, Japan
 - Developed and maintained web applications for AI-driven enterprise services, focusing on backend API implementation and integration with existing systems.
 - Implemented and tested RESTful endpoints, simple user-facing interfaces, and data-processing components in collaboration with senior engineers.

PROJECTS

- **Bayesian Optimization with Gaussian Process Panel Models (GPM)** Aug 2025 – Present
Tools: Python, Gaussian Processes, Preference-based Bayesian Optimization, Linear/Nonlinear Models
 - Studying preference-based Bayesian optimization where observations are modeled by a General Preference Embedding Model (GPM), jointly learning latent utility functions and structured observation models at University at Albany, SUNY.
 - Analyzing regret bounds and identifiability issues when combining learned embeddings with Gaussian process models for decision making.
- **Minimal-Edges Graph Neural Networks for Efficient Graph Learning** Aug 2024 – Present
Tools: Python, PyTorch, PyTorch Geometric, Graph Neural Networks, Reinforcement Learning
 - Designed a graph neural network architecture that learns task-specific sparse adjacency structures layer by layer to reduce the number of edges used in convolution at Kwansei Gakuin University.
 - Implemented an actor-critic style edge-selection module (Adjacency Generator) that optimizes graph sparsification using policy gradient methods.
- **Capstone Project in Computer Science: Methodology for Asset Operation LLM Agents** Aug. 2025 – Present
Department of Computer Science, University at Albany, SUNY
 - Developing a methodology and software framework for Asset Operation LLM Agents, combining large language models with structured task planning, tool calling, and evaluation pipelines.
 - Implementing core components such as agent prompts, trajectory logging, and automatic DAG/plan evaluation to support reproducible experimentation on industrial asset-operations benchmarks.

LEADERSHIP EXPERIENCE

- **Technical Lead, Campus Activity Recommender (HackRPI 2025)** Nov. 2025
HackRPI 2025, Rensselaer Polytechnic Institute
 - Led a small team to design and implement a campus club and event recommender web application during a 24-hour hackathon, coordinating backend design (LinUCB bandit engine, Flask APIs, database schema).
- **Team Lead, Capstone Project in Computer Science** Aug. 2025 – Present
Department of Computer Science, University at Albany, SUNY
 - Leading a multi-student team for the senior capstone software project.

SKILLS

- **Programming Languages:** Python, C, Java
- **Web Technologies & APIs:** Flask, RESTful APIs, JSON, Google Calendar API, Gemini API
- **Database Systems:** PostgreSQL (incl. pgvector/pg_trgm), SQLite
- **Data Science & Machine Learning:** PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, SciPy, matplotlib
- **DevOps & Version Control:** Git, GitHub, Docker, Docker Compose, Linux (Ubuntu/Debian), pyenv, virtualenv
- **Specialized Areas:** Graph Neural Networks (GNNs), Bayesian Optimization, Active Learning, Preference-based BO (GPM), Reinforcement Learning, Single-cell omics analysis, Gene Regulatory Network (GRN) inference, Lab-automation / protocol-aware ML
- **Mathematical & Statistical Tools:** Linear algebra, probability theory, optimization, Bayesian modeling, graphical models (pgmpy)

HONORS AND AWARDS

- **Keyence Foundation Student Scholarship** Jul 2024
Keyence Foundation (Public Interest Incorporated Foundation)
- **Gyomu Super Japan Dream Foundation Study Abroad Scholarship** Aug 2025 – Present
Public Interest Incorporated Foundation Gyomu Super Japan Dream Foundation

ADDITIONAL INFORMATION

Languages: Japanese (Native), English (Working proficiency; TOEFL iBT 79)

Scientific Interests: AI for Science and lab automation, synthetic biology and bioinformatics, graph neural networks and Bayesian optimization, single-cell omics and gene regulatory networks, active learning and preference-based optimization

Hobbies: Brazilian Jiu-Jitsu, reading books